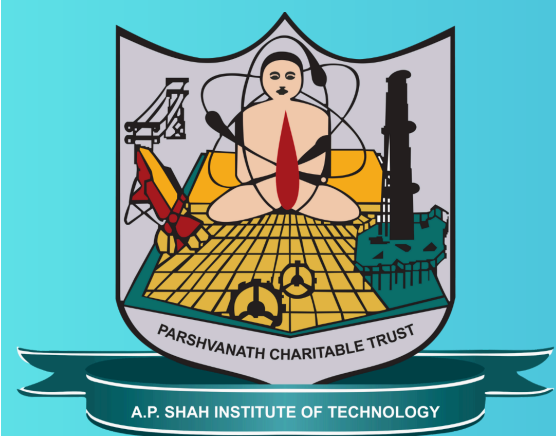
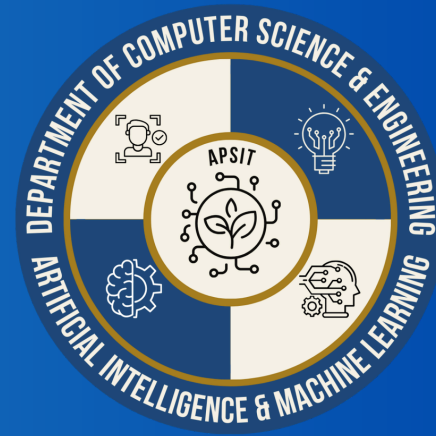




Smart Labs Zonal Control using Machine Learning

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Abstract

- Laboratories waste significant energy because lights and equipment remain active even when spaces are unoccupied, while traditional automation methods like fixed schedules and PIR sensors cannot adapt to dynamic occupancy patterns.
- This work proposes a Smart Lab Automation System using IoT, Raspberry Pi 5, and a YOLO-based computer vision model to detect human presence and enable zone-wise intelligent lighting control.
- The system processes video locally using edge computing and controls lights through GPIO-based relay modules, with a web dashboard for monitoring and manual control.

Problem Statement and Objectives

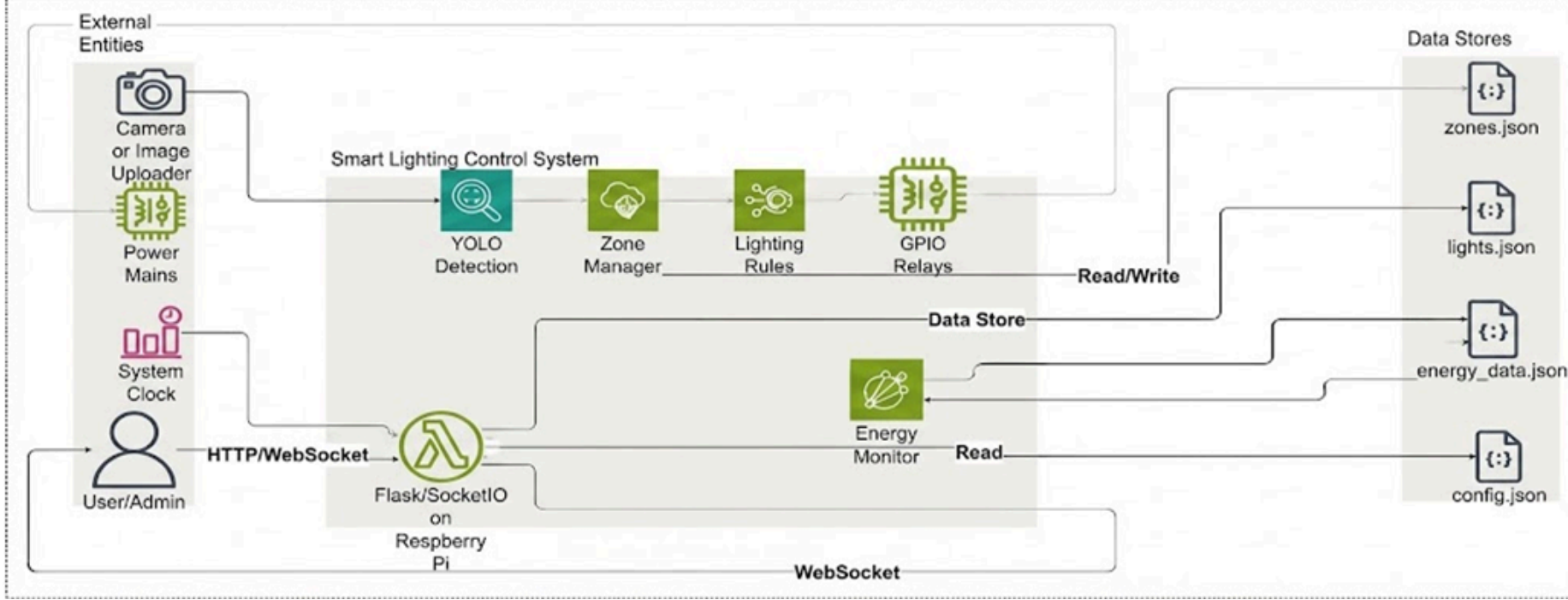
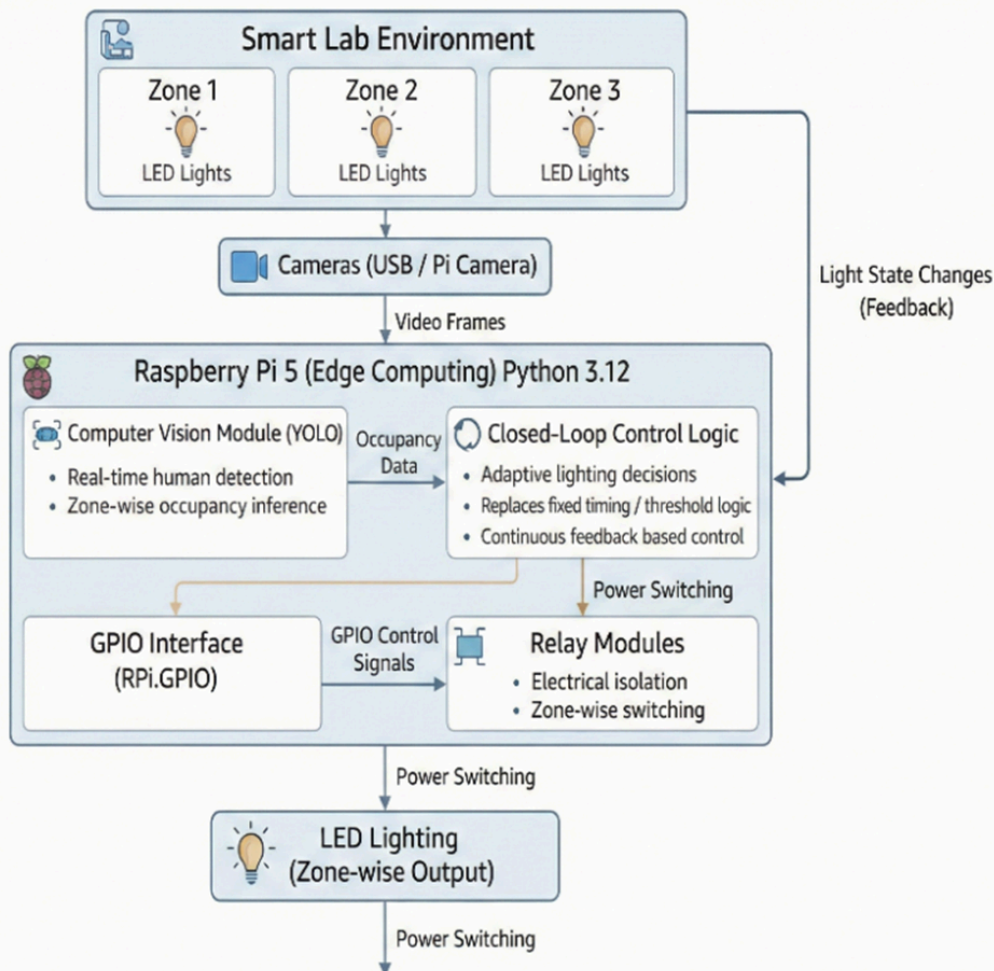
Problem Statement

- Laboratory environments waste electrical energy because lighting systems operate regardless of real-time occupancy.
- Traditional automation systems rely on fixed schedules or simple motion sensors, which cannot detect stationary occupants or zone-level usage patterns.
- Existing solutions lack intelligent decision-making and adaptive control mechanisms.

Objectives

- Design an AI-based smart laboratory automation system.
- Implement real-time human detection using YOLO.
- Enable zone-wise intelligent lighting control.
- Reduce laboratory energy consumption.
- Provide a web dashboard for monitoring and manual control.

System Design and Dataflow architecture



Results

Parameter	Conventional Lighting System	Smart Zonal Lighting System
Supply Voltage	5 V DC	5 V DC
Lighting Load	3 LEDs (1 per zone)	3 LEDs (1 per zone)
Control Method	Manual/ Always ON	Relay-based, ML-controlled
Total Daily Energy Consumption	24 Wh	9.5 Wh
Daily Energy Saving	–	14.5 Wh
Energy Consumption (10 Days)	0.72 kWh	0.285 kWh
Total Energy Saved (10 Days)	–	0.435 kWh
Percentage Energy Savings	–	≈ 60%

SDG Goals



Conclusion

- The proposed Smart Lab Automation System successfully integrates computer vision and IoT-based control to enable intelligent zone-wise lighting in laboratory environments.
- Real-time human detection using YOLO and Raspberry Pi edge computing allows lights to operate only in occupied areas, improving system efficiency.
- Experimental results show significant energy savings of about 60%, demonstrating the effectiveness of occupancy-based zonal lighting control.

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